

Monte Carlo Simulation in Vaccine Inventory Management

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Abstract— For the 2100 pandemic, this paper simulates the cost of vaccine administration in a 52-day warehouse, based on the Monte Carlo method, finding when the fixed order (y) is 3 and a fixed level stock (r) is 2, achieving the minimum total cost, which is 408.8853, namely 409 with the confidence level of 99% and the absolute error less than or equal to 1. An approximately stratified sampling approach is also used in the simulation to reduce the computation times from 35193600 to 5941200, almost six-fold reduction.

I. INTRODUCTION

The mission issued by the United Nations is to minimize the cost of storing vaccines in warehouse in the 2100 pandemic. Specifically, due to transportation constraints, a fixed-order strategy is used, namely a fixed unit of y is stocked when the storage of the warehouse is less than r units. The cost overhead of the warehouse consists of three components: a storage fee of 5 gold coins per unit, a penalty fee of 20 gold coins per day for the number of vaccines less than the number of vaccinators, and a clearance fee of 10 gold coins per unit for the last day of storage. One difficulty with the mission is that the number of people who need to be vaccinated each day is a stochastic event, so it would be an unwise to use an enumeration method [1] to iterate through each case, which not only results in a huge amount of computation but also fails to simulate the stochastic process. The Monte Carlo method [2] based on the number of features of a random distribution, the process of estimating probabilities in terms of frequencies and thus obtaining an optimal solution is well suited to this task. Another challenge is to obtain high-confidence results at low computational cost, which requires the smallest possible variance. Since the number of vaccines per day for the task is not independent, it is difficult to reduce the number of trials using traditional importance Sampling. [3] To address this difficulty, this paper adopts an approximately stratified sampling approach,[4] which first shortens the number of trial days and thus reduces the variance, and find the least-cost y and then simulate with r in the 52-day finding the minimum cost. The final result is that the running length is reduced to about one-sixth of the original number while the confidence level and the maximum value of the absolute error remain unchanged.

II. PROCESS OF COMPLETING THE MISSION

A. Choosing y and r

For the purpose of subsequent simulations, the range of a fixed unit of y and the storage level r should first be determined as well as the approximate value that may minimize the cost. Based on the table in the appendix - Probability Density of Population Visiting Vaccination Centers, the maximum number of people needing the vaccine is 6. Based on the rule in Introduction, to get the minimum cost, the amount of storage should be reduced, so the range of y is: $1 \leq y \leq 6$. And

obviously, the range of r is: $1 \leq r \leq 6$. According to the information in the appendix, the mean value of people visiting the vaccine center is 3 (round up to the nearest whole number). Based on the rule that the vaccine needs to be stocked if it is less than r , and because the number of visitors is most likely to be 3, most of the cases when y is 3 are not stocked. To maintain 3 vaccines on the next day, it is necessary to stock the vaccine every day, so r should be set to its minimum which is 1.

B. Monte Carlo simulation

Use $y = 3$ and $r = 1$ with 500 estimates, the result is shown in the following figure:

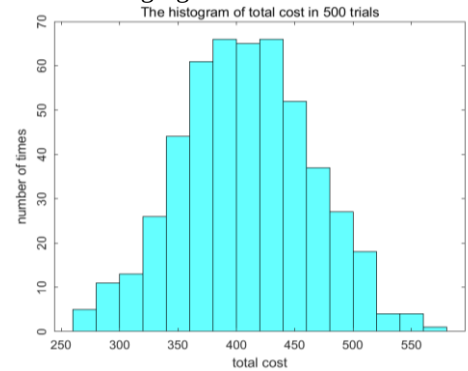


Figure1. The histogram of total cost in 500 trials with $y=3$ and $r=1$

And the sample mean is 405.85, the variance is 3156.6.

C. The actual y and r

Use the range in the A. with the Monte Carlo simulation [2], simulating 500 estimates per y and r pair. Use the sample mean [5] to represent the total cost per y and r pair, finding the minimum cost is 405.51, and $y = 3$ and $r = 2$. The y is same as the prediction in B. The reason refers to the narrative in B. The r is different from the prediction, this is because when $y = 3$, most days are no remaining vaccines, needing import vaccines everyday. And both the $r = 2$ and $r = 1$ signify the needing to import vaccines, so there is no determined prediction of it.

D. Improve the confidence

The variance [6] of the result in C. computed by Matlab is 2958.1. So the confidence interval [7] with confidence level [8] $\alpha = 99\%$ is $[399.25, 411.77]$. The width of the confidence interval is 12.52, which is more than the storage fee per unit. So it is not very exact. Therefore an increase in the number of estimations is needed to improve the accuracy of the results. Based on the mean value of the variance of the overall experiment in C., which is 2835.2, it can be estimated that the number of experiments required at a confidence level of 99%

and an absolute error [9] less than or equal to 1 (based on the exact value of gold coins in single digit) is 18,800. And the confidence interval with confidence level $\alpha = 99\%$ which is improved accuracy is [407.89,409.88].

E. Reduce the computation

The results in D., while improving accuracy to a reasonable level, rely on huge computation. Specifically it needs to be run: $52 \times 6 \times 6 \times 18800 = 35193600$ times. In order to reduce the running length, the variance needs to be reduced. It is clear that the variance decreases with the number of days in the experiment. However, the probability corresponding to numbers 5 and 6 is too small, which is 0.02, and needs to occur at least once in 50 days. In another word, if the product of the number of experiments and the number of days in the experiment is considered as an equivalent number of days, then it only needs to be 50 times in total for the numbers 5 and 6 to occur. The key is to find whether the variance [6] corresponding to a confidence level [8] of 99% and an absolute error [9] of less than 1 is satisfied for that small number of times. After more than a dozen trials, a smaller number of trials of 1050 was obtained, and the number of days was set to 2, giving the following result:

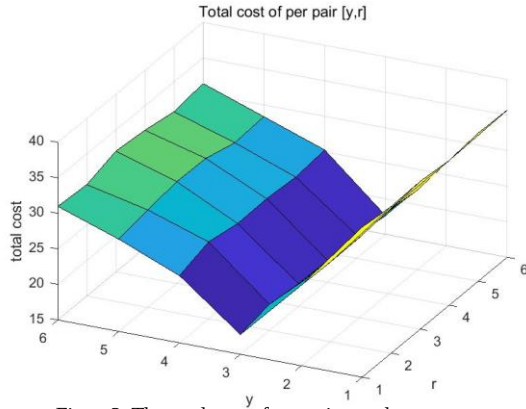


Figure2. The total cost of per pair y and r

It can be clearly observed from the above figure that the difference in total cost is greater than the preset accuracy of 1 when y is different, so the trial with 2 days is valid for y values. It is clear from Figure2. that $y = 3$ corresponds to the minimum value of cost, so re-experiment 18,800 times at $y = 3$, traversing the value of r from 1 to 6 to get the final optimal $r = 2$, which is same as the result in D. and C.. But the running length of this method reduces much, which is:

$2 \times 1050 \times 6 \times 6 + 6 \times 18800 \times 52 = 5941200$ times, almost one-sixth of the original number of tests.

III. CONCLUSION

This paper implements the Monte Carlo method of simulation to derive the optimal fixed unit y and fixed level stock r required to minimize the cost of a warehouse, accomplishing the mission of UN with a high level [8] of confidence and reasonable absolute error [9] accuracy. And an approximate stratified sampling method [4] is utilized to greatly reduce the number of experiments that would otherwise be required. In this task, I found that y seemed to have absolute dominance over the cost composition, which

also led to the fact that the difference between the costs estimated with the mean in the same y at different r was sometimes even smaller than the absolute error precision. This may be because r is supposed to be less important for the cost of the warehouse. In real life, r is critical to the cost of transportation, so perhaps the inclusion of vaccine transportation costs might be considered in the following algorithm design.

APPENDIX

TABLE 1 PROBABILITY DENSITY OF THE PEOPLE VISITING VACCINATION CENTRE

Number	Probability(P)
0	0.04
1	0.08
2	0.27324746439999997
3	0.40675253560000008
4	0.16
5	0.02
6	0.02

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REFERENCES

- [1] A Method of Defining Central and Gibbs Measures and the Ergodic Method1: A. K. Singh and S. K. Singh. A Method of Defining Central and Gibbs Measures and the Ergodic Method. Doklady Mathematics, vol. 103, pp. 72–75, 2021.
- [2] An introduction to Monte Carlo methods1: J. C. Walter and G. Barkema. An introduction to Monte Carlo methods. Physica A: Statistical Mechanics and its Applications, vol. 391, pp. 1–18, 2014.
- [3] Advances in Importance Sampling4: V. Elvira and L. Martino. Advances in Importance Sampling. arXiv preprint arXiv:2102.05407, 2021.
- [4] Area based stratified random sampling using geospatial technology in a community-based survey2: C. R. Howell et al. Area based stratified random sampling using geospatial technology in a community-based survey. BMC Public Health, vol. 20, no. 1, pp. 1678, 2020.
- [5] The Relationship Between Mean & Standard Deviation (With Example)2: Z. Statology. The Relationship Between Mean & Standard Deviation (With Example).
- [6] A new method for variance estimation in nonparametric regression2: J. Fan and I. Gijbels. A new method for variance estimation in nonparametric regression. Biometrika, vol. 85, no. 4, pp. 861–879, 1998.
- [7] Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations2: S. Greenland et al. Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations. Eur. J. Epidemiol., vol. 31, no. 4, pp. 337–350, 2016.
- [8] Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations2: S. Greenland et al. Statistical tests, P values, confidence intervals, and power: a guide to misinterpretations. Eur. J. Epidemiol., vol. 31, no. 4, pp. 337–350, 2016.
- [9] Absolute and Relative Error Calculation2: J. R. Ogden. Absolute and Relative Error Calculation. ThoughtCo, 2019.